HIGH LEVEL DESIGN (HLD)

Title

Design & Build a RAG-Based Customer Support Assistant

1. System Overview

This project implements a Retrieval-Augmented Generation (RAG) system for customer support.

Problem:

Traditional chatbots:

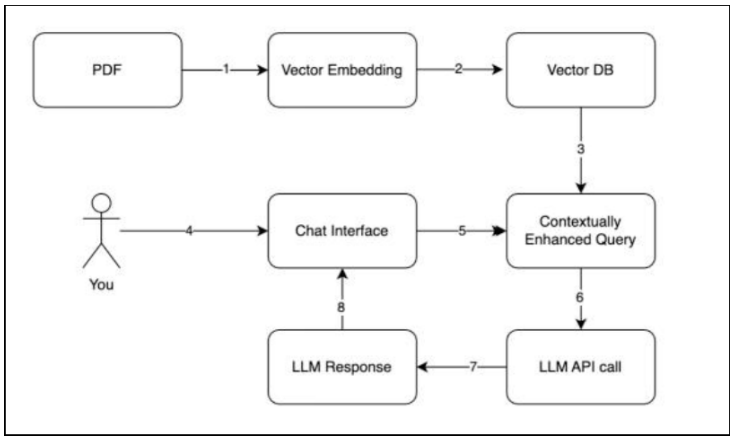
- Provide generic answers
- Do not access real documents

Solution:

A system that:

- Reads PDF knowledge base
- Retrieves relevant information
- Generates accurate responses
- Escalates complex queries

2. Architecture Diagram



3. Components Description

- User Interface → Accepts user queries
- Document Loader → Loads PDF
- Chunking Module → Splits text
- Embedding Model → Converts text to vectors
- Vector Database (ChromaDB) → Stores embeddings
- Retriever → Fetches relevant chunks
- LLM → Generates answers
- Workflow Engine (LangGraph) → Controls flow
- Routing Layer → Decides next step
- HITL Module → Handles escalation

4. Data Flow

1. Upload PDF
2. Extract text
3. Split into chunks
4. Generate embeddings
5. Store in database
6. User query received
7. Retrieve relevant chunks
8. Generate answer
9. If low confidence → escalate

5. Technology Choices

- ChromaDB → Simple and efficient vector DB
- LangGraph → Workflow control
- Sentence Transformers → Embeddings

- Python → Implementation

6. Scalability

- Efficient chunking for large documents
- Fast retrieval using vector search
- Can handle multiple users